

School of School of Science, Computing and
Engineering Technologies



Unit Outline

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COS10022

Data Science Principles

Semester January 2026

Please read this Unit Outline carefully. It includes:

PART A Unit summary

PART B Your Unit in more detail

PART C Further information



"Swinburne University of Technology recognises the historical and cultural significance of Australia's Indigenous history and the role it plays in contemporary education"

Each day in Australia, we all walk on traditional Indigenous land

We therefore acknowledge the traditional custodians of the land that our Australian campuses currently occupy, the Wurundjeri people, and pay respect to Elders past and present, including those from other areas who now reside on Wurundjeri land"

PART A: Unit Summary

Unit Code(s)	COS10022
Unit Title	Data Science Principles
Duration	One semester or equivalent
Total Contact Hours	48 hours
Requisites:	
Pre-requisites	Nil
Co-requisites	Nil
Concurrent pre-requisites	Nil
Anti-requisites	Nil
Assumed knowledge	Students are expected to have basic programming skills.
Credit Points	12.5 credit points
Campus/Location	Swinburne Vietnam (Hanoi/HoChiMinh)
Mode of Delivery	Blended
Assessment Summary	Assignment 1 (Individual, 20%) Assignment 2 (Individual, 20%) Online Test 1 (Individual, 15%) Online Test 2 (Individual, 15%) Online Test 3 (Individual, 30%)

Aims

This unit aims at introducing students to the key concepts and techniques in data science and Big Data analytics. Students will be introduced to different stages in data analytics and given the opportunity to plan and design data science projects that address business and organisational needs.

Unit Learning Outcomes

Students who successfully complete this unit can:

1. Appreciate the roles of data science and Big Data analytics in organisational contexts
2. Compare and analyse the key concepts, techniques and tools for discovering, analysing, visualising and presenting data
3. Describe the processes within the Data Analytics Lifecycle
4. Analyse organisational problems and formulate them into data science tasks
5. Evaluate suitable techniques and tools for specific data science tasks
6. Develop and execute an analytics plan for a given case study

Graduate Attributes

The Swinburne Graduate Attributes describe the capability of our graduates to use knowledge, skills and behaviours to contribute to society meaningfully and positively. They include professional, self-directed learning and future-ready skills.

This unit may contribute to the development of the following Swinburne Graduate Attributes:

- Communication 1 - Verbal communication
- Communication 2 - Communicating using different media
- Teamwork 1 - Collaboration and negotiation
- Teamwork 2 - Teamwork roles and processes
- Digital literacies 1 – Information literacy
- Digital Literacies 2 – Technical literacy

This unit contributes to the development of the following Swinburne Graduate Attributes:

- GA2 Communication - Communicating using different media: Students in this unit must use texts, images, tables, graphs or even videos to communicate findings to the audience. (Week 01, Week 07, Week 08, and Week 09, Week 11, assignment 1 and assignment 2)
- GA5 Digital literacies– Information literacy: Recognising when information is needed, locating, evaluating, and effectively using information, and responsibly creating and sharing information in various formats. Students are trained to gain skills, including critical thinking, problem-solving, research, and communication, that allow individuals to navigate the vast amount of information available in today's digital age. (Week 02, Week 07, Week 08, Week 09, Week 11, assignment 1 and assignment 2)
- GA6 Digital Literacies– Technical literacy: Understanding, using, and interacting with technology effectively. Students are trained to have the knowledge, skills, and competencies necessary to navigate and utilize various technological tools, devices, and systems in different contexts. (Week 01, Week 02, Week 03, Week 04, Week 05, Week 06, Week 10, and assignment 2).

Other graduate attributes may be practised in the unit but are not formally taught as part of the unit content, nor incorporated within formal assessment.

Content

- Introduction to Data Science and Big Data Analytics
- Roles of Data Science for Business
- Contemporary data structures and management
- Data Science tools, techniques and technologies
- The Data Analytics Lifecycle
- Analytical techniques and methods
- Analytics plan development

PART B: Your Unit in more detail

Unit Improvements

Feedback provided by previous students through the Student Survey has resulted in improvements that have been made to this unit. Recent improvements include:

- Adjusting the level of assignments and lab tasks.
- Including the content of Data Ethics and Security.
- Adjusting the question types and contents in the tests.
- Adjusting lab activities and delivery sequences.

Unit Teaching Staff

Name	Role	Email / Teams	Consultation Times
(HN) Nguyen Dinh Chi	Unit Coordinator	chinguyen@swin.edu.au	By appointment via email
(HCM) Tram Tran	Lecturer	tramtran@swin.edu.au	By appointment via email

Learning and Teaching Structure

Category	Activity	Total Hours	Hours per Week	Teaching Period Weeks
On Campus	Lectures	24 hours	2 hours	Weeks 1 to 12
On Campus	Class	24 hours	2 hours	Weeks 1 to 12

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Week by Week Schedule

Week	Week Beginning	Teaching and Learning Activity	Student Task or Assessment
1	05 Jan	▪ Lecture: Overview of Data Science and the Big Data Ecosystem. ▪ Online materials: Week 1 lecture slides. ▪ Lab: Basic Python programming and the introduction to KNIME.	▪ Readings: Chapter 1 of “Data Science for Dummies.” ▪ Readings: Chapters 1 and 2 of “Python for Data Science for Dummies.” ▪ Readings: K. Cukier and V. Mayer- Schoenberger, 2013, “The Rise of Big Data: How It's Changing the Way We Think about the World,” <i>Foreign Affairs</i>, 92(3): pp. 28-[ii]. ▪ Get familiar with the Python coding environment and the basic programming skills in Python. Get familiar with the KNIME environment.
2	12 Jan	▪ Lecture: Regression Model: Linear Regression and Evaluation/Diagnostic. ▪ Online materials: Week 2 lecture slides. ▪ Lab: Linear Regression and Logistic Regression.	▪ Study the principle of linear regression. ▪ Readings: Chapter 14 of “Data Science from Scratch. ▪ Get familiar with the Python coding environment and the basic programming skills in Python. Practice the linear regression and the logistic regression models.
3	19 Jan	▪ Lecture: Unsupervised Mode: k-Means Clustering, DBSCAN, and Evaluation/Diagnostic ▪ Online materials: Week 3 lecture slides. ▪ Lab: k-means clustering and DBSCAN.	▪ Study the concept and the classical method for unsupervised learning. ▪ Readings: Chapter 20 of “Data Science from Scratch. Practice k-means clustering and DBSCAN clustering.
4	26 Jan	▪ Lecture: Classification Mode: Naïve Bayes Classifier and Evaluation/Diagnostic. ▪ Online materials: Week 4 lecture slides. ▪ Lab: Naïve Bayes classifier.	▪ Study the principle of Naïve Bayes Classifier and the use cases. ▪ Readings: Chapter 13 of “Data Science from Scratch. Assignment 1 (individual) submission is due End of Week 04 Practice with examples of using Naïve Bayes classifier.
5	02 Feb	▪ Lecture: Ensemble Learning: Bagging, Boosting, Stacking with Decision Tree and Random Forest. ▪ Online materials: Week 5 lecture slides. ▪ Lab: Test 1	▪ Study the principle of ensemble learning and related techniques. ▪ Readings: Chapter 17 of “Data Science from Scratch. Test 1. In-class organized by lecturer
Lunar New Year Holiday from 08-Feb to 22-Feb inclusively			
6	23 Feb	▪ Lecture: Association Rules Mining ▪ Online materials: Week 6 lecture slides. ▪ Lab: Decision Tree and Random Forest classifier.	▪ Study the concept and the classical algorithm in association rules. ▪ Readings: Market Basket Analysis using R. (https://www.datacamp.com/community/tutorials/market-basket-analysis-r#firstthead) ▪ Readings: Market Basket Analysis and Recommendation Engines. (https://www.knime.com/blog/market-basket-analysis-and-recommendation-engines) ▪ Practice the decision tree classifier. Practice the Random Forest Classifier.

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Week	Week Beginning	Teaching and Learning Activity	Student Task or Assessment
7	02 Mar	▪ Lecture: Translating Business Problems into Analytic Problems, Hypothesis Formulation, and Data Store. ▪ Online materials: Week 7 lecture slides. ▪ Lab: Association Rule Mining.	▪ Readings: Chapter 3 of “Data Science for Dummies.” ▪ Readings: Chapter 3 of “Python for Data Science for Dummies.” ▪ <A data science framework advocated by DELL EMC > case study. Practice finding association rules with the topic of market basket analysis.
8	09 Mar	▪ Lecture: Data Preparation ▪ Online materials: Week 8 lecture slides. ▪ Lab: Data Preparation with Samples.	▪ Details of Data preparation processes study. ▪ Readings: Chapter 10 of “Data Science from Scratch.” Practicing data preparation procedures.
9	16 Mar	▪ Lecture: Exploratory Data Analysis and Statistical Methods for Evaluation ▪ Online materials: Week 9 lecture slides. ▪ Lab: Test 2.	▪ Study the exploratory process of data analysis and the concept of statistics. Readings: Chapters 3 and 5 of “Data Science from Scratch.” Test 2. In-class organized by lecturer
10	23 Mar	▪ Lecture: Advanced Predictive Mode: Artificial Neural Network. ▪ Online materials: Week 10 lecture slides. ▪ Lab: Introduction to Statistics and Feature Selection.	▪ Study the popular structures of artificial neural networks. ▪ Readings: Chapter 18 of “Data Science from Scratch.” ▪ Practice on the descriptive statistics and the inferential statistics. Assignment 2 (individual) submission is due End of Week 10
11	30 Mar	▪ Lecture: Communicate Results and Model Deployment. ▪ Online materials: Week 11 lecture slides. ▪ Lab: Artificial Neural Network.	▪ Study the principle of communicating results and deploying models. ▪ Readings: Chapters 26 and 27 of “Data Science from Scratch.” Practice the usage of the artificial neural network.
12	06 Apr	▪ Lecture: Revision. ▪ Online materials: Week 12 lecture slides. ▪ Lab: Test 3.	▪ Review the whole contents of this semester. Discussing questions and answers. Test 3. In-class organized by lecturer

Assessment

a) Assessment overview

Tasks and Details	Individual or Group	Weighting	Mapped Unit Learning Outcomes	Mapped Graduate Attributes	Assessment Due Date
Assignment 1	Individual	20%	4,5,6	GA5	End of Week 04
Assignment 2	Individual	20%	1,2,3	GA5	End of Week 10
Test 1	Individual	15%	4,5,6	GA5	Week 05 Class
Test 2	Individual	15%	2,3,4,5,6	GA5	Week 09 Class
Test 3	Individual	30%	1,2,3,4,5,6	GA5	Week 12 Class

Assessment Requirements	Details
b) Use of generative AI (genAI) in this unit	You are not allowed to use any AI tool to produce contents in your assignments or to answer the questions in the test. Moreover, uploading any of the material from this unit to any of the generative AI (genAI) tool such as ChatGPT, Llama, etc. are all prohibited. However, there are some scenarios you can use the generative AI tool The valid use of genAI in this unit is as follows: 1. Assist your learning in producing practice questions for you to prepare the test. 2. Assist you to correct grammar issues in the report. Nevertheless, any use of genAI must be acknowledged, with prompts and outputs included in an appendix.
c) Hurdle requirements	The basic requirement for passing this unit is to accumulate at least 50 marks in the unit. Reattempt submission is not accepted in this unit. We have the test hosted in your registered session, if you are late for the test, you can only take the remaining time to do the test before the session is closed as other units are scheduled to use the computer room. If you miss the test, a 0 mark will be assigned to the corresponding test. No alternative test or assignment can be arranged.
d) Final assessment period	There is no final exam in this unit. All tests in this unit are closed book and organised in your registered lab sessions. You will be expected to be available for your lab session to take the tests.

e) Submission requirements	Assignments and other assessments are generally submitted online through the Canvas assessment submission system which integrates with the Turnitin. Please ensure you keep a copy of all assessments that are submitted. All assignments in this unit are provided with instructions. You are expected to follow the given sequence to answer the questions and find answers. Some questions need to be answered based on the answers you found in the previous questions. If you are not following the instructed sequence, you may get incorrect answers and get deduction to your marks. Turnitin is inbuilt in Canvas submission system. You are expected to submit your assignments after passing the similarity check. The similarity value must not exceed 30% except the parts coming from the template or some short answers. If the similarity of your submission is too high, your tutor will contact you and ask you to resubmit an updated version within the given time. The assessment cover sheet should be submitted with the assignment report as a single file and pass the Turnitin check. The KNIME workflow file with the corresponding input/output files used in KNIME need to be packed as a zip file and submit to the corresponding location on Canvas. Your submission will not be marked if the zip file containing the KNIME workflow and the input/output files is missing, and a zero mark will be assigned to your corresponding assignment. In cases where a hard copy submission is required an Assessment Cover Sheet must be submitted with your assignment. The standard Assessment Cover Sheet is available from the Submitting work webpage or www.swinburne.edu.au/studentforms/
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f) Extensions and late submissions	Extensions – Extensions can only be issued according to a valid medical certificate issued by authorised GP in Melbourne. A valid medical certificate should reveal information regarding the patient and the impact dates unfit to work or study. The convenor will assess the overlap between the impact dates and days before the submission deadline to decide whether to approve the extension application and the days of extension can be given. A medical certificate can only be used once in each assessment. If you suffer from illness longer than the medical certificate states, you need to see the GP and get another valid medical certificate to apply for further extension. If the extension goes beyond the date of submitting the final result to the university, a special exam covers all contents in this unit is required for the student to take. The mark for the affected assignment will be the average mark achieved in the corresponding assignment and the special exam. Late Submissions – Unless an extension has been approved, late submissions will result in a penalty. You will be penalised 10% of your achieved mark for each working day the task is late, up to a maximum of 5 working days. The penalty is calculated by Canvas system directly. We follow whatever the system produced in the penalty result. After 5 working days, a zero result will be recorded.
g) Referencing	To avoid breaching academic integrity, you are required to provide references whenever you include information from other sources in your work and acknowledge when you have used Artificial Intelligence (AI) tools (such as ChatGPT). Further details regarding academic integrity are available in Section C of this document. Referencing conventions required for this unit are: - The prompt used to get the responses. - The responses from the AI tool. Helpful information on referencing can be found at http://www.swinburne.edu.au/library/referencing/
h) Groupwork guidelines	There is no group work in this unit.

Required Textbook(s)

The required textbook(s) can be purchased from bookshops and may be available through the Swinburne Library.

- Joel Grus, 2019. Data Science from Scratch – First Principles with Python, 2nd Edition, O'Reilly Media, Inc.
- Dietrich, D. ed., 2015. Data Science & Big Data Analytics: Discovering, Analyzing, Visualizing and Presenting Data. EMC Education Services.
- Schutt, R. and O'Neil, C., 2013. Doing data science: Straight talk from the frontline. O'Reilly Media, Inc.
- Pierson, L., Data Science for Dummies, 2nd Edition, John Wiley & Sons © 2017
- Mueller, J. P. and Massaron, L., Python for Data Science for Dummies, 2nd Edition, John Wiley & Sons © 2019 (432 pages), ISBN:9781119547624

Recommended Reading Materials

Swinburne Library has a large collection of resources. Listed below are some references that will provide valuable supplementary information to this unit. It is also recommended that you explore other sources to broaden your understanding.

- Han, J., Pei, J., & Kamber, M. (2011). Data mining: concepts and techniques. Elsevier.
- D. Lucy. (2005). Introduction to Statistics for Forensic Scientists Chichester: Wiley.
- KNIME® Beginner's Luck: A Guide to KNIME analytics Platform for Beginners by Rosaria Silipo
- KNIME® Advanced Luck: A Guide to KNIME Analytics Platform for Advanced Users by Rosaria Silipo and Jeanette Prinz
- KNIME® A Collection of Case Studies: Practicing Data Science by Rosaria Silipo
- More recommended readings are listed in the lecture slides.

PART C: FURTHER INFORMATION

For further information on any of these topics, refer to Swinburne's Student webpage <http://www.swinburne.edu.au/student/>

Student behaviour and wellbeing

All students are expected to: act with integrity, honesty and fairness; be inclusive, ethical and respectful of others; and appropriately use University resources, information, equipment and facilities. All students are expected to contribute to creating a work and study environment that is safe and free from bullying, violence, discrimination, sexual harassment, vilification and other forms of unacceptable behaviour.

The [Student Charter](#) describes what students can reasonably expect from Swinburne in order to enjoy a quality learning experience. The Charter also sets out what is expected of students with regards to your studies and the way you conduct yourself towards other people and property.

You are expected to familiarise yourself with University regulations and policies and are obliged to abide by these, including the [Student Academic Misconduct Regulations](#), [Student General Misconduct Regulations](#) and the [People, Culture and Integrity Policy](#). Any student found to be in breach of these may be subject to disciplinary processes.

Examples of expected behaviours are:

- conducting yourself in teaching areas in a manner that is professional and not disruptive to others
- following specific safety procedures in Swinburne laboratories, such as wearing appropriate footwear and safety equipment, not acting in a manner which is dangerous or disruptive (e.g. playing computer games), and not bringing in food or drink
- following emergency and evacuation procedures and following instructions given by staff/wardens in an emergency response.

Canvas

You should regularly log on to the Swinburne learning management system, Canvas. You can access Canvas via the [Student login](#) webpage or <https://swinburne.instructure.com/> Canvas is updated regularly with important unit information and communications.

Communication

All communication will be via your Swinburne email address. If you access your email through a provider other than Swinburne, then it is your responsibility to ensure that your Swinburne email is redirected to your private email address.

Academic Integrity

Academic integrity is about taking responsibility for your learning and submitting work that is honestly your own. It means acknowledging the ideas, contributions and work of others; referencing your sources and acknowledging the use of generative artificial intelligence;

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contributing fairly to group work; and completing tasks, tests and exams without cheating. Artificial intelligence tools should only be used where approved by the Unit Convenor.

Swinburne University uses the Turnitin system, which helps to identify inadequate citations, poor paraphrasing and unoriginal work in assignments that are submitted via Canvas. Your Unit Convenor will provide further details.

Plagiarism, collusion, contract cheating, unauthorised file sharing, falsification, fabrication, manipulation or misrepresentation of information, reuse of previous work and non-compliance with instructions in an invigilated or non-invigilated assessment item are all breaches of academic integrity and treated as academic misconduct. Examples of breaches of academic integrity include, but are not limited to:

- submitting work as your own for assessment that has been fully or partially completed by a third party, either paid or unpaid
- using output from artificial intelligence tools (e.g. ChatGPT) in whole or part without acknowledgement and/or without the approval of the Unit Convenor
- using another person's work or ideas as though it is your own work, without appropriate attribution
- working closely with another student or group of students (either past or current), to submit for assessment, some or all of the other student or students' work as your own work
- sharing without permission of the Unit Convenor, Swinburne resources or other material related to assessment to an entity or document repository site
- creating, intentionally modifying or inventing information that is intended to be submitted as part of an assessment item
- using the whole or part of a computer program written by another person as your own without appropriate acknowledgement
- poorly paraphrasing somebody else's work
- using a musical composition or audio, visual, graphic and photographic work created by another person without acknowledgment
- enabling others to cheat, including letting another student copy your work or by giving access to a draft or completed assignment
- letting someone or something else impersonate you, or you impersonate someone else in an invigilated or non-invigilated assessment item
- accessing, obtaining and/or providing to others unauthorised materials relating to an invigilated or non-invigilated assessment item.

The penalties for academic misconduct can be severe, ranging from a zero grade for an assessment task through to exclusion from Swinburne. For further details, see <https://www.swinburne.edu.au/student-login/academic-integrity/>

Student support

Swinburne offers a range of services and resources to help you complete your studies successfully. Your Unit Convenor or studentHQ can provide information about the study support and other services available for Swinburne students. For further information, see the [Current students](#) web page.

Special consideration

If your studies have been adversely affected due to serious and unavoidable circumstances outside of your control (e.g. severe illness or unavoidable obligation), you may be able to apply for special consideration (SPC).

Applications for Special Consideration are submitted via the SPC online tool normally no later than 5.00pm on the third working day after the submission/sitting date for the relevant assessment component. See <https://www.swinburne.edu.au/life-at-swinburne/student-support-services/special-consideration-assistance/>

Note: Submitting fraudulent (fake or altered) medical certificates is considered misconduct and can lead to serious penalties from Swinburne. In addition, your doctor may report fraudulent medical certificates as a prosecutable offence under the Victorian Crimes Act.

AccessAbility Services

If you are a student with a disability, medical or mental health condition or you have significant carer responsibilities, you may require reasonable adjustments to fully access and participate in education. Swinburne's AccessAbility Services can develop an Education Access Plan (EAP) that includes the services and reasonable adjustments that you need.

It is recommended that you register with AccessAbility Services when you first commence your course but you can contact the service at any time during your studies to find out about reasonable adjustments. Contact [Accessability Services](#) to discuss further.

Review of marks

An independent marker reviews all fail grades for major assessment tasks. In addition, a review of assessment is undertaken if your final result is between 45 and 49 or within 2 marks of any grade threshold.

You can ask the Unit Convenor to check the result for an assessment item or your final result. Your request must be made in writing within 10 working days of receiving the result. The Unit Convenor can discuss the marking criteria with you and check the aggregate marks of assessment components to identify if an error has been made. This is known as local resolution. If you are dissatisfied with the outcome of the local resolution, you can lodge a formal complaint.

Feedback, complaints and suggestions

In the first instance, discuss any issues with your Unit Convenor. If your concerns are not resolved or you would prefer not to deal with your Unit Convenor, then you can complete a feedback form. See <https://www.swinburne.edu.au/corporate/feedback/>

Advocacy

If you require assistance with any academic issues, University statutes, regulations, policies and procedures, you are advised to seek advice from an Independent Advocacy Officer at the Swinburne Student Association. Talking to an Advocacy Officer is free, independent and confidential. For more information and booking an appointment, please see <https://www.swinburne.edu.au/current-students/student-services-support/advocacy/>